Outline

Overall Project Structure (Files + Notebooks)

- `data/raw/zoopla_raw.csv` — crawler output (already collected)
- `notebooks/01_data_acquisition_and_validation.ipynb`
- `notebooks/02_data_quality_and_visual_exploration.ipynb`
- `notebooks/03_data_cleaning_and_feature_engineering.ipynb`
- `notebooks/04_annotation_and_labeling.ipynb`
- `notebooks/05_dataset_readiness_demo.ipynb`
- `documentation/dataset_card.md`, `documentation/methodology.md`, `documentation/responsible_ai.md`
- `report/group_report.docx` (final report)
- `src/` (reusable scripts for cleaning, feature engineering, imputation, modelling)

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1. Data Acquisition Strategy (10 marks)

Objective: clearly describe the problem context, data requirements, data collection process, tools used, and the justification for the chosen collection method.

Required Sections

1. Problem Context & Data Needs

- Describe the problem (e.g., analysing factors influencing housing prices and preparing a dataset for price prediction models).
- List required variables (price, area_sqft, bedrooms, bathrooms, epc_rating, city/postcode, description, property_type, etc.).

- Define quality requirements (completeness, geographic coverage, dataset size ~12k records).

2. Justification of the Data Collection Pathway

- Explain precisely that the data was collected using a browser extension (`zoopla_extension`) together with backend support (`zoopla_crawler_app/app`).
- Explain the technical reason for using an extension:
  - avoids anti-bot mechanisms
  - supports JavaScript-rendered content
  - uses browser cookies/session
  - reduces CAPTCHA triggers
- Discuss limitations:
  - slower data collection
  - potential sampling bias
  - lower automation
  - reproducibility concerns
- Explain mitigation strategies:
  - rotating human browsing sessions
  - logging metadata (time, user agent)
  - randomising navigation
  - documenting sampling criteria

3. Sources, Tools, and Processes

List the technology stack:

- Python
- requests (if used)
- BeautifulSoup (if used)
- Selenium (if used)

- browser extension code
- pandas

Include a simple flowchart:

```
Zoopla Pages
      ↓
Browser Extension Capture
      ↓
Backend Aggregator
      ↓
CSV Dataset
```

Visuals / Deliverables

Notebook:

`notebooks/01_data_acquisition_and_validation.ipynb`

Include:

- description of the data source
- screenshot or code snippet showing extension output
- metadata for `data/raw/zoopla_raw.csv`
  - number of rows
  - number of columns
  - capture date

Documentation:

- `documentation/methodology.md` (data acquisition subsection)

Acceptance Criteria

- Clear explanation of why the extension approach was used (technical justification + trade-offs).
- Raw CSV dataset available with metadata.

- Clear description allowing the dataset to be extended or recollected.

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2. Data Quality, Validation, and Visual Exploration (10 marks)

Objective: evaluate dataset completeness, consistency, and accuracy; detect bias, imbalance, and noise; and explain findings using visualisations.

Notebook:
`notebooks/02_data_quality_and_visual_exploration.ipynb`

Steps

1. Dataset Summary

Use:

```
df.info()
df.describe()
```

Create a table including:

- column names
- data types
- non-null counts
- example values

2. Missing Values Analysis

- Calculate missing percentage per column.
- Visualisations:
  - missingness heatmap
  - bar chart of missing percentages

Explain possible causes, e.g.:

- `epc_rating` missing because agents did not upload EPC certificates
- `area_sqft` missing due to differences between listing types

3. Value Consistency & Type Checks

Check:

- price formatting (currency symbols, invalid values)
- bedrooms/bathrooms numeric consistency
- EPC rating standardisation (A–G)

4. Distribution Analysis

Create plots for:

- price histogram (normal scale + log scale)
- bedrooms distribution
- bathrooms distribution
- `area_sqft` distribution
- property_type counts
- top 10 cities

5. Geographic Bias

Visualisations:

- listings per city
- boxplot of price per city
- postcode area heatmap (if available)

6. Outlier Detection

Use:

- boxplots for price and area
- IQR rule

Inspect extreme records and define a handling strategy.

7. Correlation Analysis

- correlation matrix for numeric features
- scatter plots:
  - price vs area
  - price vs bedrooms

Deliverables

Plots saved in:

```
reports/figures/
```

Tables:

```
reports/tables/data_quality_summary.csv
```

Acceptance Criteria

- All visuals included **with interpretation**, not just shown.
- Major data quality issues identified and explained.
- Suspicious or extreme records documented.

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3. Data Cleaning, Feature Engineering, and Visual Diagnostics (5 marks)

Objective: apply justified data cleaning, perform feature engineering, and validate the results using visual diagnostics.

Notebook:

`notebooks/03_data_cleaning_and_feature_engineering.ipynb`

Data Cleaning Steps

Preprocessing

- Remove currency symbols from `price`
- Convert `price` to numeric
- Standardise `area_sqft`
- Convert `epc_rating` to uppercase

Duplicate Detection

Primary key:

```
url
```

Fallback combination:

```
title + address + price
```

Missing Value Handling

Examples:

Bedrooms/bathrooms:

- median imputation grouped by property_type or city

Area:

- if missing rate ≤ 20% → predictive imputation using RandomForest or XGBoost
- if missing rate > 30% → remove rows or add missing indicator feature

EPC rating:

```
fill "Unknown"
create epc_missing_flag
```

Outlier Treatment

Options:

- winsorisation (1–99 percentile)
- log transformation

Justify with visual comparison.

Feature Engineering

Examples:

```
price_per_sqft = price / area_sqft
```

```
total_rooms = bedrooms + bathrooms + living_rooms
```

Size category:

```
small / medium / large
```

EPC mapping:

```
A → 7
G → 1
Unknown → 0
```

Text features from description:

- has_garden
- has_parking
- has_balcony
- is_renovated

If postcode extracted:

```
postcode_area
```

Visual Diagnostics

Show:

- before/after distributions
- imputation validation
- price_per_sqft distribution
- city comparison plots

Deliverables

Datasets:

```
data/cleaned/zoopla_cleaned.csv
data/processed/zoopla_features.csv
```

Scripts:

```
src/data_pipeline/impute_area.py
```

Acceptance Criteria

- All cleaning steps reproducible
- Imputation validated
- New features saved and validated

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4. Annotation and Labelling Framework (10 marks)

Objective: design a transparent and reproducible labelling framework.

Notebook:

```
notebooks/04_annotation_and_labeling.ipynb
```

Annotation Plan

Primary Label: Price Category

```
affordable  : price < £300,000
mid_range   : £300k – £800k
luxury      : price > £800k
```

Explain the reasoning for these thresholds.

Implementation

Add column:

```
price_category
```

Optional:

```
price_category_by_city
```

Quality Checks

- label distribution plot
- mean price_per_sqft per label
- random sample inspection

Subjectivity Discussion

Example:

A £500k property may be mid-range nationally but expensive in smaller cities.

Optional Additional Labels

From descriptions:

- has_garden
- has_parking

Validate using a small manually checked sample.

Deliverables

```
data/processed/zoopla_labeled.csv
documentation/annotation_rules.md
```

Acceptance Criteria

- Labels reproducible
- Visual validation performed
- Limitations documented

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5. Dataset Readiness Demonstration (5 marks)

Objective: demonstrate that the dataset can support machine learning.

Notebook:

```
notebooks/05_dataset_readiness_demo.ipynb
```

Steps

Define Task

Regression:

```
predict price
```

or classification:

```
predict price_category
```

Train/Test Split

Example:

```
80 / 20
```

Baseline Models

- Linear Regression
- RandomForestRegressor

Feature Encoding

- One-hot encoding for property_type and city
- include epc_score and text flags

Evaluation Metrics

- RMSE
- MAE
- R^2

Visuals:

- predicted vs actual scatter plot
- feature importance chart

Interpretation

Discuss which features are most predictive.

Expected strong predictors:

- area_sqft
- price_per_sqft
- bedrooms
- city

Deliverables

```
models/price_rf.pkl
```

Notebook with results and plots.

Acceptance Criteria

- Model trains successfully
- Sensible metrics reported
- Interpretation provided

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6. Responsible AI, Data Governance, and Communication (5 marks)

Documentation:

```
documentation/responsible_ai.md
```

Topics to Cover

Bias & Fairness

Examples:

- city imbalance
- property type imbalance
- missing EPC data

Privacy

Possible PII:

- exact addresses
- listing identifiers

Mitigation:

- remove or hash full addresses
- retain city and postcode_area only

Risks & Mitigation

Examples:

- location bias in price prediction
- ethical scraping concerns

Data Governance

Include:

- data dictionary
- dataset provenance
- known limitations
- intended use cases

Communication

Provide a **dataset card** summarising:

- dataset purpose
- composition
- collection method
- limitations

Deliverables

```
documentation/dataset_card.md
documentation/responsible_ai.md
```

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7. Individual Contribution and Reflection (5 marks)

Include in:

```
report/group_report.docx
documentation/individual_contributions.md
```

Each member writes **200–400 words** including:

- specific responsibilities
- challenges faced
- lessons learned
- confirmation of contributions

Extra “Outstanding” Suggestions

To make the project stronger:

- Extract **UK postcode areas** from address text using regex.
- Create **postcode-level statistics**.
- Add **map visualisations** of average prices.
- Perform **NLP analysis** on property descriptions.
- Validate imputation models using cross-validation.
- Create a **dataset card similar to “Datasheets for Datasets”**.